ref input, full runs — small-4way

L1D 16 sets x 4, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

0 of 15 runs completed, 8 running, 7 pending; **15 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
xz_cld	0/1	—	—	—	—	—	—
xz_combined	0/1	—	—	—	—	—	—
x264_pass1	0/1	—	—	—	—	—	—
gcc_pp_O3	0/1	—	—	—	—	—	—
gcc_pp_O2	0/1	—	—	—	—	—	—
gcc_smaller	0/1	—	—	—	—	—	—
gcc_ref32_O5	0/1	—	—	—	—	—	—
gcc_ref32_O3	0/1	—	—	—	—	—	—
mcf	0/1	—	—	—	—	—	—
perlbench_diffmail	0/1	—	—	—	—	—	—
perlbench_splitmail	0/1	—	—	—	—	—	—
bwaves_1	0/1	—	—	—	—	—	—
bwaves_2	0/1	—	—	—	—	—	—
bwaves_3	0/1	—	—	—	—	—	—
bwaves_4	0/1	—	—	—	—	—	—

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
xz_cld	pending	—	—	—	—	—	—	—	—
xz_combined	running	—	—	—	—	—	—	—	—
x264_pass1	running	—	—	—	—	—	—	—	—
gcc_pp_O3	pending	—	—	—	—	—	—	—	—
gcc_pp_O2	pending	—	—	—	—	—	—	—	—
gcc_smaller	pending	—	—	—	—	—	—	—	—
gcc_ref32_O5	pending	—	—	—	—	—	—	—	—
gcc_ref32_O3	pending	—	—	—	—	—	—	—	—
mcf	running	—	—	—	—	—	—	—	—
perlbench_diffmail	running	—	—	—	—	—	—	—	—
perlbench_splitmail	running	—	—	—	—	—	—	—	—
bwaves_1	pending	—	—	—	—	—	—	—	—
bwaves_2	running	—	—	—	—	—	—	—	—
bwaves_3	running	—	—	—	—	—	—	—	—
bwaves_4	running	—	—	—	—	—	—	—	—

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.